

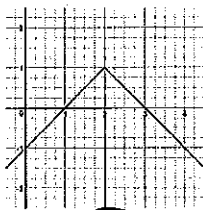
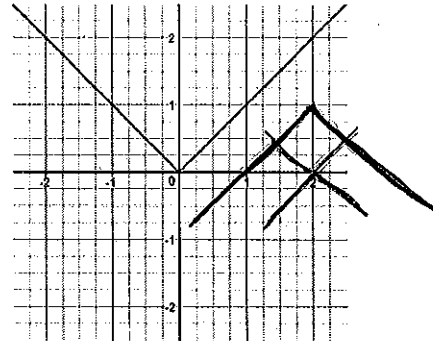
Part I - Multiple Choice - Circle your answer choice.

- 1.) Find the asymptotes of the function $f(x) = \frac{9x}{x^2 + 2x + 1}$
- HA: $y = 0$
VA: $x^2 + 2x + 1 = 0$
 $(x+1)^2 = 0$
 $x+1 = 0$
 $x = -1$
- (a) Horizontal Asymptote at $y = 9$, Vertical Asymptote at $x = 1$
(b) Horizontal Asymptote at $y = 9$, Vertical Asymptote at $x = -1$
(c) Horizontal Asymptote at $y = 9$, Vertical Asymptote at $x = 1$ and $x = -1$
(d) Horizontal Asymptote at $y = 0$, Vertical Asymptote at $x = 1$
(e) Horizontal Asymptote at $y = 0$, Vertical Asymptote at $x = -1$

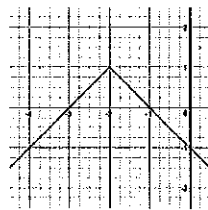
- 2.) Given the graph of $y = f(x)$, find the graph of

$y = -f(x-2) + 1$

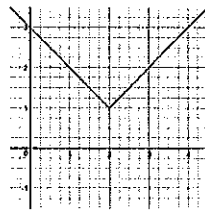
Flip over x-axis
Right + 2
up + 1



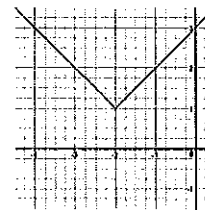
(a)



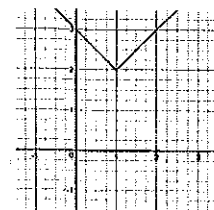
(b)



(c)



(d)



(e)

- 3.) Find the domain of the function $f(x) = \sqrt{x-1}$

$$x - 1 \geq 0$$

$$x \geq 1$$

(a) $x \leq 1$

(b) $x < 1$

(c) $x \geq 1$

(d) $x > 1$

(e) $-1 \leq x \leq 1$

- 4.) Given $f(x) = x^2$ and $g(x) = x\sqrt{x+5}$, $(f \circ g)(x)$ is equal to

$$f(g(x)) = (x \cdot \sqrt{x+5})^2 = x^2 \cdot (\sqrt{x+5})^2 = x^2 \cdot (x+5) = x^3 + 5x^2$$

(a) $x^3 + 5x^2$

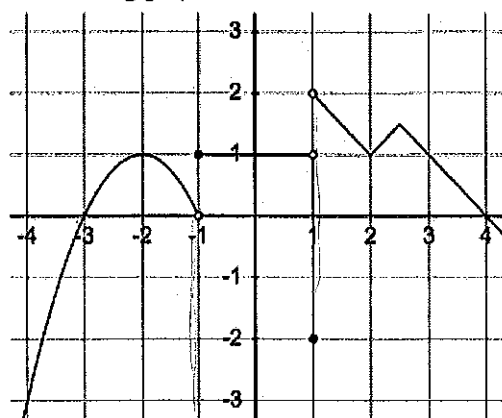
(b) $x^2 + 5x$

(c) $x^2 + (x+5)$

(d) $x^2\sqrt{x+5}$

(e) $x\sqrt{x+5}$

The following graph of the function is to be used for problems 5. – 6.



5.) Find the limit: $\lim_{x \rightarrow -1^-} f(x) = 0$

- (a) 0 (b) 1 (c) 2 (d) 3 (e) Does not exist

6.) Find the limit: $\lim_{x \rightarrow 1} f(x)$

- (a) 0 (b) 1 (c) 2 (d) 3 (e) Does not exist

7.) Find the limit: $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2} = \lim_{x \rightarrow 2} \frac{(x+3)(\cancel{x-2})}{(\cancel{x-2})} = (2) + 3 = 5$

- (a) 0 (b) 3 (c) -5 (d) 5 (e) Does not exist

8.) Find the limit: $\lim_{x \rightarrow \infty} \frac{x^2 - 5x}{2x^2 + 1} = \frac{1}{2}$

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2 (e) Does not exist

9.)

Find the limit: $\lim_{x \rightarrow 2} \frac{x+1}{3x-6} = \frac{3}{0}$

(a) 0

(b) 3

(c) $+\infty$ (d.) $-\infty$

(e.) Does not exist

10.)

Compute $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$ where $f(x) = \begin{cases} |x|, & x \leq 0 \\ \sqrt{x} + 2, & x > 0 \end{cases}$

$$\begin{aligned} &= |0| &= \sqrt{0} + 2 \\ &= 0 &= 2 \end{aligned}$$

(a) $\lim_{x \rightarrow 0^-} f(x) = 0, \quad \lim_{x \rightarrow 0^+} f(x) = 0$

(b) $\lim_{x \rightarrow 0^-} f(x) = 2, \quad \lim_{x \rightarrow 0^+} f(x) = -2$

(c) $\lim_{x \rightarrow 0^-} f(x) = -2, \quad \lim_{x \rightarrow 0^+} f(x) = 2$

(d.) $\lim_{x \rightarrow 0^-} f(x) = 2, \quad \lim_{x \rightarrow 0^+} f(x) = 0$

(e.) $\lim_{x \rightarrow 0^-} f(x) = 0, \quad \lim_{x \rightarrow 0^+} f(x) = 2$

11.) For what value of the constant k is the function $f(x)$ continuous at $x=1$.

$$f(x) = \begin{cases} kx^2 + 3x - 1, & x \leq 1 \\ \frac{k}{2} - 5x, & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

$$k(1)^2 + 3(1) - 1 = \frac{k}{2} - 5(1)$$

$$k + 2 = \frac{k}{2} - 5$$

$$k - \frac{k}{2} = -5 - 2$$

$$\frac{k}{2} = -7$$

$$k = -14$$

(a) 0

(b) 1

(c) -7

(d.) -14

(e.) No value of k makes $f(x)$ constant at $x=0$

12.) Find the slope of the secant line to the function $f(x) = x^2 + 1$ between $x = 1$ and $x = 3$

$$m = \frac{\Delta y}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{((3)^2 + 1) - ((1)^2 + 1)}{2} = \frac{10 - 2}{2} = 4$$

- (a) 1 (b) 2 (c) 3 (d.) 4 (e.) 5

13.) The Equation of the tangent line to the curve $f(x) = x^2 + 4x - 2$ where $x = 2$ is

i.) Find Slope: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^2 + 4(x+h) - 2] - [x^2 + 4x - 2]}{h}$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - 2 - x^2 - 4x + 2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2 + 4h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 4)}{h} = \lim_{h \rightarrow 0} 2x + h + 4 = 2x + 4$$

$$m = f'(2) = 2(2) + 4 = 8$$

ii) Find Point: $y_0 = f(2) = (2)^2 + 4(2) - 2 = 10$

iii) Find Eqn:

$$y - y_0 = m(x - x_0)$$

$$y - 10 = 8(x - 2)$$

$$y - 10 = 8x - 16$$

$$y = 8x - 6$$

- (a) $y = 8x - 6$ (b) $y = 8x + 6$ (c) $y = 2x + 2$ (d) $y = 2x - 2$ (e) $y = 2x + 8$

14.) For what value of the constant k is the function $f(x)$ continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{\sin(8x)}{x}, & x < 0 \\ k, & x \geq 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{x \rightarrow 0^-} 8 \cdot \frac{\sin(8x)}{8x} = \lim_{x \rightarrow 0^+} k$$

$$8 \cdot \lim_{x \rightarrow 0^-} \frac{\sin(8x)}{8x} = k$$

$$8 = k$$

- (a) 0 (b) $\frac{1}{8}$ (c) 1 (d.) 8 (e.) No value of k makes $f(x)$ constant at $x = 0$

15.)

Compute $A = \lim_{x \rightarrow +\infty} \frac{x^2 + x}{x + 1}$ and $B = \lim_{x \rightarrow -\infty} \frac{x^2 + x}{x^2 + 1}$ (Note: The two functions are not the same.)

$$= -\infty$$

$$= \lim_{x \rightarrow -\infty} \left(\frac{x^2 + x}{x^2 + 1} \right) \cdot \frac{\left(\frac{1}{x^2} \right)}{\left(\frac{1}{x^2} \right)} = \lim_{x \rightarrow -\infty} \frac{1 + \cancel{\frac{1}{x}} \rightarrow 0}{1 + \cancel{\frac{1}{x^2}} \rightarrow 0}$$

$$= 1$$

(a.) $A = +\infty, B = 1$ (b) $A = 1, B = +\infty$ (c) $A = -\infty, B = 1$ d.) $A = +\infty, B = +\infty$ e.) $A = 1, B = 1$

16.)

Find the limit: $\lim_{\beta \rightarrow 0} \frac{\sin(\beta)}{\beta}$

(a) 0

(b) 1

(c) $-\infty$ d.) $+\infty$

e.) Does not exist

17.)

Suppose that $\lim_{x \rightarrow 2} f(x) = 1$ and $\lim_{x \rightarrow 2} g(x) = 5$. Then $\lim_{x \rightarrow 2} \frac{2f(x)g(x)}{f(x) + g(x)}$ is equal to

$$= \frac{2 \cdot (1) \cdot (5)}{(1) + (5)} = \frac{10}{6} = \frac{5}{3}$$

(a) $-\frac{5}{3}$

(b) -1

(c) 1

(d.) $\frac{5}{3}$

e.) 15

- 18.) Compute the average rate of change of the function $f(x) = x^2 + 1$ on the interval $[2, 4]$

$$\frac{f(4) - f(2)}{4 - 2} = \frac{((4)^2 + 1) - ((2)^2 + 1)}{2} = \frac{17 - 5}{2} = \frac{12}{2} = 6$$

- (a) 12 (b) 3 (c) 5 (d) 8 (e) 6

- 19.) A ball is thrown straight up into the air with an initial velocity of 50 ft/sec. The height $h(t)$ of the ball after t seconds is given by $h(t) = 50t - 16t^2$. What is the average velocity for the time period beginning when $t = 1$ and lasting 0.1 seconds?

$$\frac{h(1.1) - h(1)}{1.1 - 1} = \frac{(50(1.1) - 16(1.1)^2) - (50(1) - 16(1)^2)}{0.1}$$

$$= \frac{35.64 - 34}{0.1} = 16.4 \text{ ft/sec}$$

- (a) 18 ft/sec (b) 50 ft/sec (c) 16.4 ft/sec (d) 1.64 ft/sec (e) 32 ft/sec

- 20.) Find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for the given function $f(x) = x^2 - 2x + 15$

$$\frac{[(x+h)^2 - 2(x+h) + 15] - [x^2 - 2x + 15]}{h}$$

$$= \frac{x^2 + 2xh + h^2 - 2x - 2h + 15 - x^2 + 2x - 15}{h} = \frac{2xh + h^2 - 2h}{h}$$

$$= \frac{h(2x + h - 2)}{h} = 2x + h - 2$$

- (a) $2x + h$ (b) $2x + 2h - 2$ (c) $2h + x$ (d) $2h + x - 2$ (e) $2x + h - 2$

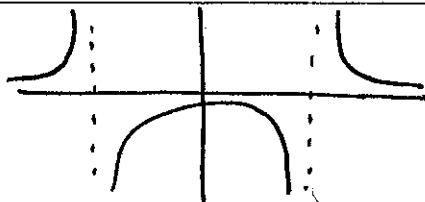
Part II - Free Response - You must show ALL your work in the space provided to receive credit.

Calculate the limit (by manipulation/evaluation) if it exists.

21. $\lim_{x \rightarrow 1} x^2 - 2x + 1 = (1)^2 - 2(1) + 1$

0

22. $\lim_{x \rightarrow 3} \frac{x+4}{x^2-x-6} = \frac{7}{0}$



DNE

23. $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 12}{x^2 - 9} = \lim_{x \rightarrow 3} \frac{(x-4)(x-3)}{(x+3)(x-3)} = \frac{3-4}{3+3} = \frac{-1}{6}$

$-\frac{1}{6}$

24. $\lim_{x \rightarrow 0} \frac{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}{x(\sqrt{x+1}+1)} = \lim_{x \rightarrow 0} \frac{x+1-x}{x(\sqrt{x+1}+1)}$
 $= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+1}+1} = \frac{1}{\sqrt{1}+1} = \frac{1}{2}$

$\frac{1}{2}$

25. $\lim_{x \rightarrow \infty} \frac{-3x^2 + x}{x^2 + 4x - 1} = \frac{-3}{1}$

-3

$$26. \lim_{x \rightarrow \infty} \frac{\sqrt{16x^2 - 2x + 6}}{2x - 1} = \frac{\sqrt{\frac{1}{x^2}}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{16 - \frac{2}{x} + \frac{6}{x^2}}}{2 - \frac{1}{x}} = \frac{\sqrt{16}}{2} = \frac{4}{2} = 2$$

2

$$27. \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 4} - x)(\sqrt{x^2 + 4} + x)}{\sqrt{x^2 + 4} + x} = \lim_{x \rightarrow \infty} \frac{x^2 + 4 - x^2}{\sqrt{x^2 + 4} + x} = \lim_{x \rightarrow \infty} \frac{4}{\sqrt{x^2 + 4} + x} = \frac{4}{\infty + \infty} = \frac{4}{\infty} = 0$$

0

28. Given that $f(x) = \begin{cases} 2x, & x < 1 \\ 2, & x = 1 \\ x^2 + 1, & x > 1 \end{cases}$, find

a. $\lim_{x \rightarrow 1^+} f(x) = (1)^2 + 1 = 2$

b. $\lim_{x \rightarrow 1^-} f(x) = 2(1) = 2$

c. $\lim_{x \rightarrow 1} f(x) = 2$

d. $f(1) = 2$

29. Is the function continuous at $x = 1$? Yes.

Why or why not? i.) $\lim_{x \rightarrow 1} f(x)$ exist, ii.) $f(1)$ exist. iii.) $\lim_{x \rightarrow 1} f(x) = f(1)$

30. State the definition of a function $f(x)$ which is continuous at $x = a$:

i.) $\lim_{x \rightarrow a} f(x)$ exist.

ii) $f(a)$ exist.

iii) $\lim_{x \rightarrow a} f(x) = f(a)$

31. Given $f(x) = x^2 + 5x - 4$,

a. Find the slope of the tangent line at the point $x=2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 5(x+h) - 4 - [x^2 + 5x - 4]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 5x + 5h - 4 - x^2 - 5x + 4}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 5h}{h} = \lim_{h \rightarrow 0} \frac{\cancel{h}(2x + h + 5)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 2x + \cancel{h} + 5 = 2x + 5$$

$$m = f'(2) = 2(2) + 5 = 9$$

9

b. Find the equation of the tangent line at $x=2$

$$y_0 = f(2) = (2)^2 + 5(2) - 4 = 10$$

$$y - y_0 = m(x - x_0)$$

$$y - 10 = 9(x - 2)$$

$$y - 10 = 9x - 18$$

$$y = 9x - 8$$

$$y = 9x - 8$$

32.

Use the definition of derivative to find the derivative function of $f(x) = \frac{5}{3x}$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\left(\frac{5 \cdot x}{3(x+h) \cdot x} - \frac{5 \cdot (x+h)}{3x \cdot (x+h)} \right)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\left(\frac{5x - 5(x+h)}{3(x+h) \cdot x} \right)}{h} = \lim_{h \rightarrow 0} \frac{\frac{5x - 5x - 5h}{3(x+h) \cdot x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{-5 \cdot \cancel{h}}{3(x+h) \cdot x} \cdot \frac{1}{\cancel{h}} = \lim_{h \rightarrow 0} \frac{-5}{3(x+\underset{\downarrow 0}{h}) \cdot x} \\
 &= \frac{-5}{3 \cdot x \cdot x} = -\frac{5}{3x^2}
 \end{aligned}$$

$$f'(x) = -\frac{5}{3x^2}$$

33.

Use the definition of derivative to find the derivative function of $f(x) = 2\sqrt{x} - 1$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(2\sqrt{x+h} - 1) - (2\sqrt{x} - 1)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} - \cancel{1} - 2\sqrt{x} + \cancel{1}}{h} = 2 \cdot \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\
 &= 2 \cdot \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h \cdot (\sqrt{x+h} + \sqrt{x})} = 2 \cdot \lim_{h \rightarrow 0} \frac{\cancel{x+h} - x}{\cancel{h}(\sqrt{x+h} + \sqrt{x})} \\
 &= 2 \cdot \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+\underset{\downarrow 0}{h}} + \sqrt{x}} = \frac{2}{\sqrt{x} + \sqrt{x}} = \frac{2}{2 \cdot \sqrt{x}} \\
 &= \frac{1}{\sqrt{x}}
 \end{aligned}$$

$$f'(x) = \frac{1}{\sqrt{x}}$$